AeroVis - Mixed Reality Wire Harness

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***Abstract* - Overview: AeroVis aims to revolutionize the design process and maintenance in the aviation industry through the implementation of mixed reality. This is done by converting a schematic into a binary map and using the A* algorithm to optimize the pathfinding of the components in the harness.**

Methods: Utilizing the Unity Engine in order to develop a Mixed Reality (MR) application. The application is loaded onto a Microsoft HoloLens2 for use in Mixed Reality. The application is developed using C# and Python inside of the Unity Engine v2022.3.45f1. Using a user-uploaded schematic, a python script will convert that image into grayscale. This grayscale image will be used to make a binary map of the schematic. This binary map will use an A* algorithm in order to route out the most efficient paths for the wire harness and display the result to the user.

Keywords - Mixed Reality, Unity Development, HoloLens2, A* Algorithm

I. Introduction

A. The Problem

The problem that AeroVis seeks to solve is a problem that every individual in any manufacturing, design, or engineering field has experienced before: the requirement to walk away from our work to look at the schematics. AeroVis seeks to solve that problem by implementing the use of Mixed Reality into the manufacturing process through the eyes of Microsoft's HoloLens2 Mixed Reality Headset. The aim of the project is to display an accurate, to scale, schematic that can be seen and interacted with while the user is still able to engage with the world around them in real time.

B. Literature Search

To begin work on AeroVis when it was pivotal to search for projects that correspond to the topic at hand and see if something similar had been done that could aid[1]. Using the foundation set by the resulting sources to not only gain a better understanding of their task but also gather insights into how others approached similar challenges and how AeroVis can build upon that pre-existing work to achieve the penultimate objective. The search results highlighted diverse applications of advanced algorithms and mixed reality (MR). They include improving helicopter safety with the city based A* Algorithm [1], enhancing industrial training with MR [2], and using MR glasses for robotic pathfinding [3]. Research also covers object-centered UIs in MR [4], interactive learning tools for aircraft engineering [5], and personalized RV design with HoloLens 2 [6]. Additionally, advancements in the A* Algorithm and its application in game pathfinding are discussed, showing improved performance in various scenarios.

C. Personality Tests

In order to build team chemistry from the get-go, it was decided to see how the group's personality types aligned. Using Carl Jung's and Isabel Briggs Myers' typology, it was concluded that Ryan resulted in a typology of INFJ (Introverted, iNtuitive, Feeling, Judging); and Rudy had resulted in a type of ESTJ (Extroverted, Sensing, Thinking, Judging). These results showcased how as a group the personality difference can actually compliment the other. This promoted covering a large spectrum of the personality typology and interacting with as large of an audience as possible.

D. Complex Engineering Problem

The group decided that a current problem plaguing the STEM world currently is the usage and training of Artificial Intelligence in the workplace and commercially. The big focus currently is regarding the training data, the limitations of when and when not to use artificial intelligence, and how it is vital to monitor the exponential knowledge growth of these systems. How can leaders in the computer science and engineering field track the knowledge of these systems and potentially have access to prevent these systems from surpassing the human brain when the time comes? Engineering concerns include how humans would keep up with artificial intelligence when it inevitably surpasses human capabilities. Science issues arise when artificial intelligence is coming up with their own breakthroughs and as humans need to prove the complex solutions these computers will create; even if we do not

1. Literature search was performed on September 5, 2024 in correspondence with Sacred Heart University's Librarian Kristina D'Agostino

understand. This is also true for mathematics since proofs are one of the hardest mathematical standards to meet.

E. Engineering Solution

In order to plot out the potential directions the project can go, our project began with brainstorming solutions to reach the goal of our project. By creating a box diagram displaying what features would be ideal in the implementation of the application. For the project this year one of the main areas of interest is one centralized user-interface (UI). This would make navigation and operation of the application itself more efficient, and make it look more presentable; regardless of the finished project. Since the project previously was without a universal solution to the problems at hand, anysort of foundation that will be set in place now will be set firmly. For the project going forward, rather than building off of what was previously completed, it would be better to use their work and their failures to determine points of most significance. Analyzing these failures, along with the successes alongside them, will result in an outline of the most important topics to tackle and take on early. Most importantly, implementation of a central interface that would guide the user around the application and streamline the schematic process would make the project fit better as an application. It would also be beneficial to potentially have calibration settings for all potential users saved and be able to be called back for accessibility purposes.

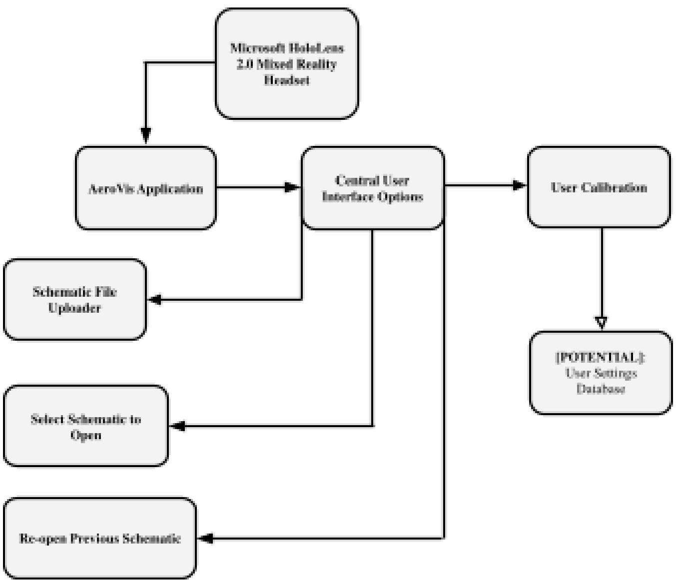

Figure 1: Box diagram displaying the initial foundation and plans for the project direction.

F. Team Charter

Team Charter

Mixed Reality (MR) Aircraft Wiring Harness Assembly

Purpose

- This project's purpose is to integrate Mixed Reality (MR) technology into the aircraft wiring harness assembly process to enhance efficiency, accuracy, and reliability.

Goals

- Develop a hardware and software interface for compatibility between existing assembly tools.
- Ensure high accuracy and reliability of MR overlays in varying conditions.
- Optimize the layout and routing of wiring harnesses to minimize material usage and assembly time.

Team Members and Roles

- **Rudy Ferrera**: Develop hardware and software interfaces and ensure compatibility with MR technology.
- **Ryan Monast**: Develop hardware and software interfaces and ensure compatibility with MR technology.

Timeline and Milestones

- **Phase 1**: Research and Planning - Identify existing tools and techniques and assess MR technology requirements.

Meeting Schedule

- **Team Time**: Every Thursday at 5:30 PM.
- **Mentor Meetings**: Every Tuesday at 11:00 AM.

Figure 2: The team charter used in outlining roles and goals for the team during the duration of this project.

G. Meeting Minutes

Week Three

- **Time:** 11:30am
- **Discussion:** Our discussion this week centered around planning the upcoming semester and making sure our goals were not only attainable but reasonable. This discussion also included our mentors.
- **Attendees:** Ryan, Rudy, Michael, and Ryan
- **Means of Communication:** Met in person.
- **Advancements Made:** We have set definitive landmark dates for throughout the semester, and after meeting with our mentors have plans for when we want to meet during the rest of the semester.
- **Goals for Next Week:** Gantt Chart, Progress Report #1 Completed, Progress Report Presentation #1 Completed, and a functional HUD to demo for the mentors next week.
- **Other:** We currently sit one week ahead of our schedule and hope to use this momentum to succeed early on.

Figure 3: How meeting documentation was handled in the weekly meetings for the project.

H. First Semester Progression

The first weeks of the project were spent studying the desired functions, design specifications, and previous foundations from the group before us. In the third week of the project, the group made the first alpha version of the program available on GitHub for the project. This first version included very rudimentary features including basic hand tracking and early control layouts. This milestone was also monumental because the program was able to be successfully uploaded and tested on the HoloLens, which was a concern for the group last year. Building off of this application, within a week the project had nearly doubled in scope. This was in large part due to the material in the first three reference sources containing similar content to what was being worked on at that time. This second version was now equipped with a fully functional hand menu and schematic visualization. However, the next few weeks were spent bug fixing the problems from the second version. These bugs included head tracking issues, schematic and hand menu clipping errors, and LiDAR issues with the canvas display. During the debugging process it was also important to the project to have a physical demonstration, since only one person can be in the HoloLens at once. The group decided that Rudy, using his experience in C coding, 3D printing, and arduino development, would work primarily on the physical demonstration. This in turn meant Ryan would be doing the work on the software side of the project, since he knows more about C++, Unity, and had a better computer to handle the expensive rendering requirements for an application such as AeroVis. In the more recent weeks, seeing the previous successes in the earlier versions, the mentors of the group thought it would be an amazing idea to implement a barcode scanner, which would allow the user to not have to move the schematic one step at a time. While Ryan worked to implement this idea using the recommended program, Vuforia, Rudy caught the demonstration up to where the Unity software is just in time for the third release. This release included early Vuforia editions, button labeling updates, and canvas updates.

I. Looking Forward

The next steps in the project will be vital in the overall completion of our goal. This is because in the coming weeks the project will largely shift to implementing the bulk of the scripting, since a majority of the work before this point has been UI focused. The physical demonstration will now take a back seat as the Unity application looks to move into a Beta phase.

II. References

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